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1 Chapter Name

1.1 Axioms for $\mathbb{R}$

Problem 1

1. If $\rho : F \rightarrow G$ is an isomorphism of fields, then the inverse mapping $\rho^{-1} : G \rightarrow F$ is also an isomorphism. Thus $F \cong G \implies G \cong F$.
2. If $\rho : F \rightarrow G$ is a monomorphism of fields, then $\rho(1) = 1$.
3. The field $\mathbb{Q}$ of rational numbers is not isomorphic to the field $\mathbb{Q}(t)$ of rational forms over $\mathbb{Q}$.

Proof. Suppose $\rho : F \rightarrow G$ is an isomorphism of fields. Let $a, b \in G$. Since ρ is bijective, there exists unique $x, y \in F$ such that $\rho(x) = a, \rho(y) = b$, so $\rho^{-1}(a) = x, \rho^{-1}(b) = y$. Now

$$\rho^{-1}(a + b) = \rho^{-1}(\rho(x) + \rho(y)) = (\rho^{-1} \circ \rho)(x + y) = x + y = \rho^{-1}(a) + \rho^{-1}(b).$$

Similarly,

$$\rho^{-1}(ab) = \rho^{-1}(\rho(x)\rho(y)) = (\rho^{-1} \circ \rho)(xy) = xy = \rho^{-1}(a)\rho^{-1}(b).$$

Thus ρ^{-1} is a homomorphism, and it is bijective since ρ is, so ρ^{-1} is an isomorphism. ■

Proof. Suppose $\rho : F \rightarrow G$ is a monomorphism of fields. Since ρ is injective and $\rho(0) = 0$, we have $\rho(1) \neq 0$. Then

$$1^2 = 1 \iff \rho(1^2) = \rho(1) \iff \rho(1)\rho(1) = \rho(1).$$

Cancelling $\rho(1) \neq 0$ gives $\rho(1) = 1$. ■

Proof. Suppose $\rho : \mathbb{Q} \rightarrow \mathbb{Q}(t)$ is a monomorphism. By (ii) we have $\rho(1) = 1$, then $\rho(n) = n$ for all $n \in \mathbb{Z}_{>0}$ by induction. Since $\rho(-1) + \rho(1) = \rho(0) = 0$, we have $\rho(-1) = -1$, so $\rho(n) = n$ for all $n \in \mathbb{Z}$. For $p/q \in \mathbb{Q}$, we have $q \cdot (p/q) = p$, so $\rho(q)\rho(p/q) = \rho(p)$, giving $\rho(p/q) = p/q$. Thus $\rho(x) = x$ for all $x \in \mathbb{Q}$, so ρ is not surjective, a contradiction. ■

Problem 2

In the field $F = \mathbb{Q} + i\mathbb{Q}$ of Gaussian rationals, if $\alpha = r + is$ define $\bar{\alpha} = r - is$. Show that the mapping $\rho : F \rightarrow F$ defined $\rho(\alpha) = \bar{\alpha}$ is an automorphism of F , and that, for every nonzero element α , $\alpha^{-1} = (r^2 + s^2)^{-1}\bar{\alpha}$.

Proof. Let $\alpha, \beta \in F$ and suppose $\alpha = a + ib, \beta = x + iy$. Then

$$\rho(\alpha + \beta) = \rho((a + x) + i(b + y)) = (a + x) - i(b + y) = a - ib + x - iy = \rho(\alpha) + \rho(\beta).$$

Similarly,

$$\rho(\alpha\beta) = \rho((a + ib)(x + iy)) = \rho((ax - by) + i(ay + xb)) = (ax - by) - i(ay + xb) = \rho(\alpha)\rho(\beta).$$

Since $\rho(\rho(\alpha)) = \bar{\bar{\alpha}} = \alpha$, ρ is its own inverse and thus bijective, so ρ is an automorphism. For the second part, we verify directly

$$\alpha \cdot (r^2 + s^2)^{-1}\bar{\alpha} = (r + is)(r - is)(r^2 + s^2)^{-1} = (r^2 + s^2)(r^2 + s^2)^{-1} = 1.$$

Thus $\alpha^{-1} = (r^2 + s^2)^{-1}\bar{\alpha}$. ■

1.2 The Order Axioms

Problem 1

In an ordered field, $0 < a < b \iff b^{-1} < a^{-1}$.

Proof. Suppose $0 < a < b$.

$$a^{-1} - b^{-1} = a^{-1}(b - a)b^{-1}.$$

On the rhs, $b - a > 0$ by hypothesis. Since $b > a > 0$, by Theorem 1.2.3 (12) we have $a^{-1} > 0$ and $b^{-1} > 0$. Then by O_2 the product on the rhs is in P , thus $b^{-1} < a^{-1}$.

Conversely, suppose $b^{-1} < a^{-1}$. If $a \leq 0$ then $a^{-1} \leq 0$, and if $b \leq 0$ then $b^{-1} \leq 0$, so $b^{-1} < a^{-1} \leq 0$, contradicting $b^{-1} < a^{-1}$. Thus $a > 0$ and $b > 0$. If $0 < b < a$, then by the forward direction $a^{-1} < b^{-1}$, contradicting $b^{-1} < a^{-1}$. Thus $0 < a < b$. ■

Problem 2

In an ordered field, $a^2 + b^2 = 0 \implies a = b = 0$. Similarly, for a sum of n squares ($n = 3, 4, 5, \dots$).

Proof. Suppose $a^2 + b^2 = 0$. Suppose for contradiction that $a \neq 0$ or $b \neq 0$, WLOG $a \neq 0$. Then $a^2 > 0$ by Theorem 1.2.3 (9) and $b^2 \geq 0$, so $a^2 + b^2 > 0$ by O_1 . This contradicts $a^2 + b^2 = 0$. Thus $a = b = 0$. For a sum of n squares $x_1^2 + \dots + x_n^2 \geq x_1^2 \geq 0$. But $x_1^2 + \dots + x_n^2 = 0$ thus $x_1 = 0$ and for each x_i it follows similarly. ■

Problem 3

In an ordered field, $a > b > c \geq 0$ and $a \leq b + c$, then

$$\frac{a}{1+a} \leq \frac{b}{1+b} + \frac{c}{1+c}.$$

Proof. Suppose $a > b > c \geq 0$ and $a \leq b + c$. Then

$$\begin{aligned} \frac{b}{1+b} + \frac{c}{1+c} - \frac{a}{1+a} &= \frac{b(1+a)(1+c) + c(1+b)(1+a) - a(1+b)(1+c)}{(1+a)(1+b)(1+c)} \\ &= \frac{b + ab + bc + abc + c + ac + bc + abc - a - ab - ac - abc}{(1+a)(1+b)(1+c)} \\ &= \frac{b + c - a + 2bc + abc}{(1+a)(1+b)(1+c)} \\ &= \frac{(b + c - a) + bc(2 + a)}{(1+a)(1+b)(1+c)}. \end{aligned}$$

Since $a \leq b + c$ we have $b + c - a \geq 0$, and since $a > b > c \geq 0$ all of b, c, a are non-negative so $bc(2 + a) \geq 0$. Thus the numerator is ≥ 0 and the denominator is > 0 , giving

$$\frac{b}{1+b} + \frac{c}{1+c} - \frac{a}{1+a} \geq 0. \quad \blacksquare$$

Problem 5

Let F be an ordered field, a and b any elements of F , and n an odd positive integer. Prove:

1. $a < b \iff a^n < b^n$;
2. $a = b \iff a^n = b^n$;
3. $a > b \iff a^n > b^n$;

Proof. (i) Suppose $a < b$. If $a \geq 0$ we have Theorem 1.2.8. Thus suppose $a < 0$. There are three cases. Suppose $b = 0$. Since n is odd $a^n < 0$ thus $a^n < b^n$. Suppose $b > 0$. Since n is odd $a^n < 0$ thus $a^n < b^n$. Suppose $b < 0$. We have $-a > -b$ and by Theorem 1.2.8 $(-a)^n > (-b)^n \iff (-1)^n a^n > (-1)^n b^n$. Since n is odd $(-1)^n = -1$ so we have $-(a^n) > -(b^n)$. Multiplying by -1 gives $a^n < b^n$. The converse follows by trichotomy. If $a^n < b^n$ then $a = b$ is impossible and $a > b$ is impossible so $a < b$.

(ii) is trivial. (iii) is (i) with the roles of a and b reversed. $\blacksquare$

Problem 6

Let $\mathbb{Q}(t)$ be the field of rational forms over $\mathbb{Q}$ and let S be the set of all nonzero elements $r(t) = p(t)/q(t)$ of $\mathbb{Q}(t)$ such that the leading coefficients of the polynomials p and q have the same sign, that is, such that $p(t) = a_0 + a_1t + \cdots + a_m t^m$, $q(t) = b_0 + b_1t + \cdots + b_n t^n$ with $a_m b_n > 0$.

1. $\mathbb{Q}(t)$ is an ordered field, with S as a set of positive elements.
2. For the ordering defined in (i), $n1 < t$ for every positive integer n .

Proof. Let $f, g \in \mathbb{Q}(t)$. Write $f = \frac{p_1}{q_1}$, $g = \frac{p_2}{q_2}$ where $p_1, p_2, q_1, q_2 \in \mathbb{Q}[t]$, with leading coefficients a_m, c_k of p_1, p_2 and b_n, d_ℓ of q_1, q_2 respectively.

(i) We verify the three axioms for S to be a set of positive elements.

Trichotomy. Let $f \in \mathbb{Q}(t)$, $f \neq 0$. Then $f = p_1/q_1$ with $p_1 \neq 0$, so $a_m b_n \neq 0$. Either $a_m b_n > 0$, in which case $f \in S$, or $a_m b_n < 0$, in which case $(-a_m)b_n > 0$ so $-f \in S$. These are mutually exclusive since $a_m b_n > 0$ and $a_m b_n < 0$ cannot both hold.

Closure under addition. Suppose $f, g \in S$, so $a_m b_n > 0$ and $c_k d_\ell > 0$. Then

$$f + g = \frac{p_1}{q_1} + \frac{p_2}{q_2} = \frac{p_1 q_2 + p_2 q_1}{q_1 q_2}.$$

The leading coefficient of $q_1 q_2$ is $b_n d_\ell$. The leading coefficient of $p_1 q_2 + p_2 q_1$ is determined by the highest-degree term; since $\deg(p_1 q_2) = m + \ell$ and $\deg(p_2 q_1) = k + n$, if $m + \ell \neq k + n$ the leading coefficient is $a_m d_\ell$ or $c_k b_n$ respectively, both of which are positive (as products of same-sign pairs); if $m + \ell = k + n$ the leading coefficient is $a_m d_\ell + c_k b_n > 0$. In all cases the numerator's leading coefficient times $b_n d_\ell > 0$, so $f + g \in S$.

Closure under multiplication. Suppose $f, g \in S$, so $a_m b_n > 0$ and $c_k d_\ell > 0$. Then

$$fg = \frac{p_1 p_2}{q_1 q_2}.$$

The leading coefficient of $p_1 p_2$ is $a_m c_k$ and of $q_1 q_2$ is $b_n d_\ell$. Then $(a_m c_k)(b_n d_\ell) = (a_m b_n)(c_k d_\ell) > 0$, so $fg \in S$.

Hence S makes $\mathbb{Q}(t)$ an ordered field.

(ii) Fix a positive integer n . We must show $t - n \cdot 1 \in S$, i.e. $t - n \in S$ (identifying $n \cdot 1$ with the constant polynomial n). We have $t - n = \frac{t-n}{1}$. The leading coefficient of the numerator $t - n$ is $1 > 0$ and the leading coefficient of the denominator 1 is $1 > 0$, so $(1)(1) > 0$ and thus $t - n \in S$, giving $n < t$. ■